Basics about Artificial Intelligence

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Abstract

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I. AI Literacy Fundamentals

A. What is AI?

Artificial Intelligence (AI) is any technique that enables computers to mimic human intelligence, such as for example recognising speech, understanding language, solving problems, making decisions, or learning from data. Sub-sets of AI include Generative AI, Machine Learning, Deep Learning.

1. Machine Learning

Machine learning is a subset of Artificial Intelligence that includes techniques that enable machines to improve at tasks with experience. By analysing large datasets, machine learning algorithms can identify patterns and make predictions or decisions based on data.

This technology powers many everyday applications such as personalised recommendations on streaming platforms, spam email filters, and fraud detection systems. Machine learning forms the foundation of many AI advancements, including natural language processing, image recognition, and predictive analytics.

Another key AI technology is Deep Learning. Deep Learning is a subset of Machine Learning based on neural networks that permits a machine to train itself to perform a task.

2. Generative AI

Generative AI (Gen AI) refers to a subset of AI that focuses on creating new content, such as text, images, videos, or code, by learning patterns from existing data. Unlike Machine Learning systems that are mostly designed to classify or predict, Gen AI generates original output based on user inputs or prompts. Popular examples include tools such as ChatGPT, Gemini, and Claude.

Examples include drafting text, generating visuals, content creation, summarising documents, and modelling scenarios. While the adoption of Gen AI continues to grow, its effectiveness

and limitations vary depending on the context and the tasks it is applied to. We will see how later in the course. Popular examples include tools such as ChatGPT, Gemini, and Claude for text and computer code, and tools such as Midjourney and Dall-E for images.

Generative AI is increasingly being applied across a wide range of industries and roles, offering tools to support productivity and creativity.

B. Technologies of AI

AI includes a variety of technologies that enable machines to perform tasks that typically require human intelligence.

In this section, we will cover some of the most popular and relevant functions for you to know about such as supervised and unsupervised learning, which allow systems to learn from data.

We will also cover large language models (LLMs) which focus on understanding and generating human language.

Let's explore these key AI technologies, their mechanisms, and their real-world applications across diverse fields.

1. Supervised Learning

Supervised learning is a type of machine learning where the machine is taught using examples with known answers. In other words, the machine is given input data along with the correct output, so it can learn how to map inputs to output.

Imagine teaching a baby to recognise a dog. You show the baby a picture of a dog and say, "This is a dog." Over time, the baby learns to identify dogs by seeing more examples.

Supervised learning works in a similar way: the machine is shown input data (like a picture) and the corresponding output (e.g., "dog") to learn the relationship between them.

To work, supervised learning requires a training set, which is a collection of data points where each input is paired with its correct output. By analysing this labeled data, the machine learns to predict the correct output for new input.

For example:

• Input: A picture of an animal.

• Output: The label ”dog.”

Over time, the machine can identify dogs in new pictures by applying what it has learned from the training data.

2. Applications of Supervised Learning

Supervised learning is particularly useful for solving problems such as:

• Classification: Assigns categories or labels to input data, making it suitable for tasks involving distinct groups. For example, sorting a dataset based on demographic data such as sex or age.

• Regression: Predicts a value based on input data. For instance, estimating real estate prices by considering factors like size and location, or detecting email spam or suspected fraudulent transactions based on historical data patterns.

• Object Detection: Identifies and determines the location of multiple objects within an image or video. This is commonly used in scenarios where knowing the exact position of objects is critical, such as in autonomous vehicles, which detect and locate pedestrians and obstacles in their environment.

3. Unsupervised Learning

Unsupervised learning is a type of machine learning where the system learns patterns and relationships in data without being given explicit labels or predefined outcomes. Unlike supervised learning, where input data is paired with known outputs, unsupervised learning works with unlabelled data, meaning the machine is essentially ”learning by itself.”

In unsupervised learning, the system analyses raw data to uncover hidden patterns or groupings without human intervention. This makes it especially useful for situations where there are no clear answers or where subject matter experts are unsure about the properties of a dataset. By recognising similarities and differences within the data, unsupervised learning can provide valuable insights.

4. Applications of Unsupervised Learning

Unsupervised learning is particularly useful for solving problems such as:

• Clustering: Groups similar data points together based on shared characteristics, without using predefined labels. For example, it can be used in marketing to segment customers with similar purchasing behaviors into distinct groups, or to automatically organise news articles into topics without prior knowledge of the categories.

• Association: Identifies relationships or patterns between variables in a dataset, often represented as association rules that highlight co-occurrence or dependencies between them. For example it can be used by e-commerce platforms to identify items that are frequently bought together, and make recommendations.

5. Large Language Models (LLMs)

Large language models (LLMs), are advanced AI systems designed to understand language and generate text in response to a user prompt. LLMs are trained on large datasets containing text enabling them to learn patterns, grammar, and the context of words and sentences.

These models typically rely on models known as transformers, which allow them to process large amounts of text efficiently and understand relationships between words across sentences and paragraphs.

There are two main approaches to language modelling: statistical models and neural models.

• Statistical Models: These models rely on statistical methods to predict the probability of word sequences. By analysing patterns in large datasets, they estimate which words are most likely to appear next in a given context. For example, statistical models might calculate that ”sunny” is more likely to follow ”The weather is” than ”delicious.”

• Neural Models: Neural models leverage neural networks inspired by the structure of the human brain. These models excel at understanding complex relationships in language and performing advanced tasks, such as sentiment analysis, machine translation, and question answering. Modern neural models, like transformers, power state-of-the-art systems in natural language processing (NLP) by efficiently handling vast amounts of data and

Module 1: AI Literacy Fundamentals Key Takeaways

A Framework to Apply AI

Classification	Characteristic	Example	Main Outcome
Automate	**Streamline tasks:** AI tools use machine learning capabilities to automate tasks that are repetitive and standard in nature by learning from data patterns	• Automate generation of content and grading • Create AI-powered tutoring chatbots	**Drive Efficiency**
Discover	**Data Visualisation:** AI tools able to reveal patterns, trends, and relationships within the data, facilitating exploration and highlighting actionable insights	• Provide insights on the learning patterns of students • Recognise students' wellbeing and performance	**Provide Insights**
Personalise	**Dynamic Profiles:** AI tools gather data from various sources and build a dynamic user profile that evolves as user interacts with the system and as their preferences change	• Customise learning content and paths • Give personalised real-time feedback	**Boost Engagement**
Predict	**Forecasting Trends:** AI tools go beyond describing past or present data, and are able to analyse identified patterns and trends to make educated guesses of the future	• Identify at-risk students for early intervention • Predict jobs demand and skills for better employability	**Provide Insights**
Include	**Accessibility:** AI tools designed to support inclusivity and accessibility by offering dedicated interfaces and interactions that cater to varying levels of skills and abilities	• Improve accessibility to students with disabilities • Provide real-time language translation	**Augment Learning**

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capturing context across sentences.

6. Applications of Large Language Models

Large language models are widely used in areas including but not limited to:

• Text Generation: Creating human-like text for applications like writing emails, drafting articles, or generating creative content.

• Language Translation: Converting text from one language to another with high accuracy, as seen in tools such as Google Translate.

• Sentiment Analysis: Determining the emotions or opinions expressed in text, valuable for analysing customer feedback or social media sentiment.

• Chatbots: Powering conversational AI systems for customer service or intelligence assistants in natural language.

C. A Framework to Apply AI

1. AI for Automation

It is not infrequent to find ourselves spending a significant portion of our time on repetitive and administrative tasks, leaving less room for strategic or creative work. AI can play a key role in automating such tasks, enabling students and professionals to focus on higher-value activities.

For instance, in customer service, AI-powered chatbots can handle routine inquiries, such as order tracking or FAQs, providing instant responses.

In Insurance, AI can automate administrative tasks such as processing claims or scheduling appointments.

In manufacturing, AI-driven robotics can streamline assembly lines by performing repetitive tasks with precision and consistency.

These innovations not only reduce workloads but can also improve efficiency, accuracy, and customer satisfaction across various sectors.

For students, AI tools can automate tasks like transcribing lectures, generating study guides, or organising notes.

2. AI to Discover Insights

AI is a powerful tool for uncovering trends and insights that were previously difficult or even impossible to identify. For example, in healthcare, AI can analyse patient data to detect early warning signs of diseases. In marketing, AI can identify emerging consumer trends by analysing vast amounts of data, helping companies adjust their strategies to meet changing customer demands.

For students, AI can assist in discovering learning patterns, such as identifying subjects they struggle with or recognising study habits that lead to better performance. Similarly, AI can be used to understand emotional and behavioural patterns, enabling a more supportive learning experience.

3. AI for Personalisation

Many businesses face the challenge of catering to diverse needs. Personalisation powered by AI is shaping industries by tailoring experiences, products, and services to individual needs and preferences.

In retail, platforms such as Amazon and Spotify use AI to provide customised recommendations, whether it's suggesting the next purchase or creating playlists tailored to listening habits. Similarly, in finance, robo-advisors use AI to design investment portfolios aligned with a client's financial goals and risk tolerance.

For students, AI has the potential to offer a personalised learning experience by adapting to their unique strengths and challenges. For example, AI can adapt lessons based on a student's pace and accuracy, ensuring they progress effectively.

4. AI to Predict Future Trends

AI's ability to analyse past data and make predictions can bring great benefits to individuals and businesses supporting better decision-making and proactive interventions.

For example, in manufacturing, predictive analytics can help monitor machinery and predict maintenance needs, reducing downtime and improving efficiency. In transportation, AI predicts traffic patterns to optimise delivery routes and reduce delays, benefiting logistics and ride-sharing services. Similarly, in energy, AI can forecast demand and supply fluctuations, allowing utilities to manage resources efficiently and reduce costs.

In the context of career search, AI is able to perform continuous real time screening of jobs and skills databases to profile and match young graduates to the most relevant and suitable careers for them. This can allow professors to adjust and tailor content as they go and students to potentially better select subjects and focus in the course of their studies.

5. AI for Inclusivity

AI is proving to be a powerful tool for promoting accessibility and inclusion across industries.

In customer service, AI chatbots and voice assistants enable companies to support users

in multiple languages and assist individuals with disabilities.

In the workplace, AI tools such as Microsoft's Immersive Reader make documents accessible for employees with dyslexia or visual impairments. Similarly, in media and entertainment, platforms such as YouTube and Netflix use AI-driven captioning and audio description to enhance accessibility for viewers.

D. Impact of AI on Jobs and Society

1. Definition of AI-First Companies

- Distinguished from companies that just add AI features.
- An AI-first company would not exist without AI; AI is central to its core product value (e.g., OpenAI, Anthropic).

2. Three key traits

- AI is foundational to the business model, not just for efficiency gains.
- Data architecture is optimized for AI models to learn and improve continuously through customer feedback loops.
- The product evolves and improves at a very rapid pace.

3. Impact of AI on Jobs

- A major shift is expected.
- Jobs being automated: Basic, repetitive, rules-based, data-heavy tasks (e.g., basic customer support, data entry, simple coding).
- Jobs being augmented (copilot approach): Roles where AI assists but humans make final decisions (e.g., doctors for diagnosis, lawyers, financial analysts, software engineers for complex architecture).
- New jobs being created: Roles like prompt engineering, AI ethics, and data privacy management.
- Overall Outlook: Believes AI will create a net positive impact, leading to a better job environment. Short-term adjustments will occur, but it will free people from mundane tasks

to focus on higher-level work.

4. Future of Global Hiring & Workforce

• Technology (e.g., Zoom, portfolio company Oyster) enables a global, distributed workforce and access to opportunities worldwide.

• Advice for young graduates: While remote work is viable, in-person interaction is valuable early in a career for informal learning and skill development.

5. Advice for Students in an AI-Powered World

• Become AI-Literate: Despite the field's newness, anyone can become knowledgeable. Follow industry news, understand key players, and actively test new AI tools (like OpenAI's deep research feature) to understand their capabilities and limitations.

• Develop Critical Thinking: This skill is essential to evaluate the outputs of AI models. Consume diverse content (fiction, non-fiction) to continually refine this ability.

• Enter the Job Market with Informed Opinions: Employers are also learning. Students who can demonstrate tested knowledge and opinions on AI tools will be more valuable.

E. Limitations of AI and Ethical Considerations

AI widespread adoption comes with several limitations and ethical challenges that must be addressed to ensure its responsible use.

This section explores some of the critical concerns surrounding AI, focusing on key issues every student and professional should be aware of.

1. Bias and Discrimination

AI systems often inherit biases present in the data they are trained on. For instance, if historical data reflects societal inequalities, AI models can perpetuate or even amplify these biases. This becomes particularly problematic in sensitive areas like hiring, lending, or university admissions.

To mitigate bias, it is crucial to diversify the teams building AI models, regularly audit datasets, and ensure human oversight in decision-making processes.

2. Privacy and Data Ownership

AI's reliance on large datasets raises critical questions about privacy and data ownership.

Traditionally, data is often transferred to centralised entities, such as private companies, which take ownership and control of the information. This creates concerns about accountability, misuse, and loss of individual rights.

To address these issues, emerging approaches such as federated learning and decentralised data analytics are gaining traction.

Federated learning allows AI systems to train on data stored across different individuals, controllers, or jurisdictions without requiring the data to be centralised. This type of infrastructure can allow individuals to maintain control over their data while still enabling AI to perform analytics and generate insights. For instance, healthcare organisations could use federated learning to collaboratively train AI models on patient data without compromising privacy or transferring ownership to external companies.

3. Over-Reliance and Loss of Human Autonomy

AI has the potential to automate tasks, but over-reliance on AI systems can reduce human autonomy and critical decision-making. For instance, students using AI-driven content creation tools might inadvertently depend too much on the technology, risking a loss of creativity. Similarly, in workplaces, excessive automation could make employees less skilled at tasks traditionally performed by humans.

To address this, organisations should view AI as a tool to complement human effort rather than replace it. Maintaining a "human-in-the-loop" approach ensures that humans retain oversight and control over AI systems.

4. Transparency and Accountability

One of the most critical challenges with AI is determining accountability for its actions. If an AI system makes an error, who is responsible? A number of companies in highly regulated industries like finance or education, require designated individuals to take responsibility for AI-driven decisions, ensuring accountability at every level.

Embedding explainability into AI systems helps address this challenge, making it clear how decisions are made and enabling human oversight. Accountability should ideally be proportionate to both the likelihood of something going wrong and the severity of the potential consequences.

II. Best Practices for Prompting

A. Types of Prompts

When working with AI tools such as ChatGPT and Gemini, the quality and effectiveness of the responses depends largely on how you design your prompts. In other words, prompt engineering is the art of asking the right question to get the best output from an LLM.

Prompts can take various forms, each suited to specific tasks. In this chapter, we explore key types of prompts and how to use them effectively.

Informational: Gather facts or summaries

Instructional: Step-by-step tasks

Creative: Generate imaginative ideas or content

Role Assignment: Simulate a specific role or perspective

Contextual: Provide background for tailored responses

Chain-of-thought: Encourage step-by-step reasoning for complex tasks

Iterative: Refine responses with follow-up questions or feedback

B. Informational prompts

Purpose: To gather factual information, summaries, or definitions.

Examples:

•" Define the concept of sustainability in 100 words."

•" List three major causes of climate change."

Best practices and tips:

• Be concise but specific about the scope.

• Include the format you want: "List," "Summarise," "Explain briefly."

• Don't take a good result for granted. Gen AI can make mistakes and often the first iteration of the response can be easily improved.

C. Instructional prompts

Purpose: To guide the AI through a specific task or set of instructions.

Examples:

• "Create a step-by-step guide for writing a persuasive essay."

• "Generate a checklist for preparing a research paper."

Best practices and tips:

• Break tasks into smaller steps for clarity.

• Use action verbs like "Create," "Design," "Generate," or "Outline."

D. Creative prompts

Purpose: To generate imaginative content or brainstorm ideas.

Examples:

• "Write a dialogue between a time traveler and a historian debating the consequences of altering historical events"

• "Suggest three innovative business ideas for a startup in the fintech space"

Best practices and tips:

• Add creative constraints to shape the output, such as specific themes, requirements, tones, or word limits.

E. Role-assignment prompts

Purpose: To make AI simulate a specific role or perspective.

Examples:

• "Pretend you are a historian. Explain the impact of the Industrial Revolution."

• "Act as a fitness coach and create a 5-day workout plan for beginners."

Best practices and tips:

• Clearly specify the role and the task.

• Use "Act as," "Pretend to be," or "Imagine you are" to guide the response.

F. Contextual prompts

Purpose: To include background information or audience details for better responses.

Examples:

• Without context: "Explain what bitcoin is."

• With context: "Explain what bitcoin is to my grandmother"

Best practices and tips:

• Add audience, purpose, and desired tone for tailored responses.

G. Chain-of-thought prompts

Purpose: To guide the AI step-by-step through a task by asking it to explain its reasoning or thought process. This approach helps with complex questions that require logical thinking.

Examples:

• Prompt: "A train travels 60 miles in 1.5 hours. If it continues at the same speed, how far will it travel in 4 hours? Explain step-by-step."

• Expected Chain-of-Thought Response:

"First, calculate the speed of the train. To do this, divide the distance traveled by the time: 60 miles ÷ 1.5 hours = 40 miles per hour."

"Next, use this speed to calculate the distance traveled in 4 hours: 40 miles per hour × 4 hours = 160 miles."

"So, the train will travel 160 miles in 4 hours at the same speed."

Best practices and tips:

• The chain-of-thought approach ensures the AI explains the reasoning process, not just the final answer.

• Useful for students learning problem-solving techniques or needing detailed explanations.

• This approach also makes it easier to spot potential mistakes or areas of improvement

H. Iterative prompts

Purpose: To refine the output by asking follow-up questions or providing feedback to improve the response. This method allows you to tweak the prompt based on the initial results.

Examples:

• Initial Prompt: "Summarise the history of the Roman Empire."

• Follow-Up 1: "Focus only on the key events leading to its decline."

• Follow-Up 2: "Make the summary more concise."

Best practices and tips:

• Start with a broad prompt, then narrow down or adjust based on the output.

• Provide specific feedback, such as "Use simpler language" or "Add more examples."

• If something it's not clear or doesn't sound right to you, ask the AI to explain it or clarify. It will help you evaluate the effectiveness of the output.

I. Elements of a Good Prompt

Clarity: Ambiguous or vague prompts can confuse the AI, leading to unclear or irrelevant responses. Clear prompts ensure the AI understands exactly what you're asking.

Specificity: Specific prompts narrow down the scope, ensuring the AI provides a focussed and accurate response.

Context: Adding context improves the AI's understanding of your needs, making the response more relevant.

Instructions for formatting: Clearly stating how you want the output formatted helps you get usable responses.

Examples in the prompt: Providing examples gives the AI a reference point, improving the desired relevance and quality of the output.

Iteration and Refinement: Even the best prompts might need tweaking. Iterative refinement helps you achieve the desired result.

III. Evaluating AI Output

A. What to Look for in AI Output

AI technologies, whether general-purpose tools or advanced Generative AI models, require careful evaluation to ensure their reliability, fairness, and appropriateness.

This module introduces a comprehensive framework for assessing AI applications, followed by a focused examination of evaluating output from Gen AI tools.

B. Key Evaluation Criteria

A structured way to critically evaluate the output of AI tools is by focusing on the following evaluation criteria. Note that these criteria can be applied to a broad range of AI applications.

1. Data and model validity

Assess the quality and reliability of the data and algorithms used to train and test the AI model.

2. Accuracy

Assess how accurate the information generated is, with human judgment and verifiable sources.

3. Ethical and legal considerations

Ensure compliance with regulations and ethical guidelines.

4. Explainability and transparency

Ensure humans are able understand the logic behind the AI's decisions and its decision-making process.

5. Bias

Evaluate the inclusivity of training data and verify the absence of discriminatory output.

C. Key Evaluation Criteria for Gen AI Output

1. Relevance: Determine if the response directly addresses the prompt or task.
What to look for:

- Does the response stay on topic?
- Does it align with the specifics of the question?

2. Accuracy: Ensure factual correctness and verify information against credible sources.
What to look for:

- Are the facts correct?
- Are cited statistics or claims verifiable?

3. Completeness: Check if the response covers all necessary points without omitting critical details.
What to look for:

- Does the response address all parts of the prompt?
- Are important aspects or nuances missing?

4. Context: Ensure the response aligns with the intended audience, purpose, and situation.
What to look for:

- Is the tone appropriate for the audience?
- Does it fit the purpose (e.g., informative, persuasive)?

5. Bias detection: Identify and address biased or discriminatory content.
What to look for:

- Are stereotypes perpetuated in the response?
- Does the output exclude certain groups or perspectives?

6. Tone and style: Ensure the response matches the intended audience and context.

What to look for:

- Does the tone suit the task (e.g., professional, casual, persuasive)?

Is the language appropriate for the target audience?

7. Hallucinations: Detect fabricated or false information that may sound credible but lacks evidence.

What to look for:

- Are there statements that seem plausible but cannot be verified?
- Does the response include made-up facts or references?

D. Fact-Checking AI Output

Step 1: Review the initial output

- Read through the AI-generated content carefully.
- Identify statements that require verification, such as statistics, dates, or specific claims.

Step 2: Note down listed facts

- Identify key facts or claims: Highlight or list all factual statements, statistics, and dates provided by Gen AI.

Step 3: Assess source relevance

- If Gen AI provides references or links, confirm:
  - That the sources are accessible and reputable.
  - The cited information is correctly interpreted and in context.

- If no sources are provided, independently seek reliable references.

Step 4: Check factual accuracy

- Verify with reliable sources:
  - Use trusted sources such as government publications, academic articles, or credible media outlets to cross-check the information.
  - Prioritise sources with strong editorial oversight, such as scholarly databases, official websites, and industry publications by well-renowned institutions.
- Search online for supporting evidence using search engines and tools.

Step 5: Consult subject matter experts if needed

- For highly specialised or critical topics you may want to:
  - Reach out to experts in the field for verification.
  - Use academic networks or professional associations to find knowledgeable individuals.

E. Context Awareness

Step 1: Define the context

- Identify the audience: Who will read or use this information? (e.g., fellow students, your boss, clients, general public).
- Determine the purpose: What is the output intended to achieve? (e.g., inform, persuade, educate, support decision-making).
- Clarify the situation: What is the setting or medium for the response? (e.g., a classroom discussion, email to a client, business proposal, public report).

Step 2: Evaluate alignment
Assess if the response:

- Uses appropriate tone and language for the audience (e.g., formal for business setting, conversational for fellow students).

- Focuses on the relevant details that fulfill the purpose.
- Addresses the situation effectively (e.g., includes references for academic contexts, actionable insights for decision-makers).

Step 3: Identify gaps
Highlight any inconsistencies or misalignments:

- Is the language too complex or too simplistic for the audience?
- Is the tone a good fit for you and your academic / business setting?
- Does the response include unnecessary information that dilutes the purpose?
- Are key situational requirements, such as citations, missing?

Step 4: Refine for context
Modify the response to:

- Match the audience's knowledge level, interests, and expectations.
- Clearly address the intended purpose.
- Fit the specific medium or setting.

Step 5: Test and iterate
Self-review the refined response by asking yourself:

- Does it clearly address the intended audience's needs?
- Does it fulfil the purpose (e.g., inform, persuade, or support decision-making)?
- Is it relevant and appropriate for the specific situation (e.g., formal report, casual discussion)?

Simulate the audience's perspective by:

- Reading the response critically as if you are the intended recipient.
- Considering questions or objections the audience might raise.
- Make adjustments based on your review to ensure clarity, relevance, and effectiveness.

IV. Critical Thinking and Creativity in the Age of AI

A. The Role of Humans in AI

1. The Evolving Role of Humans with AI Integration

• We are in the early, foundational stages of AI technology, with its full impact and applications still unknown.

• The potential for AI to drive innovation, economic opportunity, improved public services, and solve complex challenges is tremendous.

• Soft skills are critically important and sometimes overlooked. As most people will be ordinary users, not programmers, foundational competencies like critical thinking and how we relate to information are key.

• AI integration into work is already happening (citing 2024 IMS and WEF statistics on workforce impact).

2. Critical Gaps in AI Literacy

• Policymakers & Institutions: Need upskilling to create forward-thinking, well-informed policies that support AI innovation and ensure access to high-quality, representative data.

• Education: Students are often already familiar with AI tools. The greater challenge is equipping educators to:

• Integrate AI for personalized learning, curriculum creation, and fostering creativity.

• Understand the future skills students will need for their careers.

3. Balancing AI Skills with Human Skills

• The balance between ”hard” technical skills (e.g., prompt engineering, development) and ”soft” skills (e.g., critical thinking, creativity) is dynamic and will evolve with the technology.

• Example - AI Provenance: Highlights the need for both technical solutions (industry collaboration on watermarking, metadata) and evolved human critical thinking to discern AI-generated content as it becomes more sophisticated and human-like.

• Need for both: A universal foundation of critical thinking is essential, alongside fostering technical skills to ensure AI development is locally grounded, culturally nuanced, and representative.

4. Preparing for an AI-Powered Future

• Develop Fundamental Competencies: Understand how the technology works, its limitations, and how it can complement your career or learning journey.

• Engage with AI Tools Actively: Don't shy away. Use them to foster creativity, augment learning, synthesize information, brainstorm, and improve efficiency in day-to-day tasks (e.g., copy editing, idea distillation).

• Help Shape the Future: Students can demonstrate the value of AI tools to educators and participate in discussions about classroom policies, ethics, and standards. The future needs people immersed in the technology to help shape its ethical and contextual application.

• Embrace Continuous Evolution: Recognize that jobs and career paths will change. New roles will emerge. Focus on adaptability and the ability to leverage AI as a complementary tool.

B. AI to Augment the Creative Process

1. What is Creative Thinking?

Creative thinking is the ability to approach problems or ideas in novel ways, often by synthesising diverse perspectives, generating original concepts, or reimagining existing solutions. It involves curiosity, imagination, and critical thinking, and it's central to innovation across disciplines.

Key traits of creative thinking:

• Originality: Developing new ideas or approaches.

• Problem-Solving: Generating solutions that may not be immediately obvious.

• Experimentation: Exploring multiple paths and perspectives to find the best fit.

2. How AI Impacts Creative Thinking

1. Boosting creative thinking through automation

Gen AI automates repetitive or low-level tasks, freeing up time for creators to focus on high-level conceptual work. This reduction in cognitive load can allow people to direct their mental energy toward ideation and experimentation.

2. AI as a thought partner

Gen AI can serve as a brainstorming collaborator, helping creators explore novel perspectives or generate unexpected combinations of ideas. It stimulates imagination by offering suggestions that challenge conventional thinking.

3. Challenges to creative thinking

However, the reliance on AI can hinder the development of deep creative skills. Many students and professionals use AI tools to generate content without engaging in critical thinking or adding their own insights. This "passive creativity" risks reducing the originality and depth of their work.

3. Finding the Balance: Human-AI Collaboration

To fully realise AI's potential as a creative tool, it must augment — not replace — human thinking.

Effective use of AI involves:

1. Critical engagement: Actively reviewing, refining, and building upon AI output.

2. Imaginative exploration: Using AI as a stimulus for creativity, not a substitute for it.

3. Ethical awareness: Ensuring AI-generated content aligns with originality and societal values.

C. Applying Critical Thinking in the Context of AI

1. The Role of Critical Thinking in AI Applications

Critical thinking involves the active process of questioning, analysing, and evaluating information. The methodologies we have explored in the previous lessons showed how critical thinking is essential to evaluate AI output.

To recap: Critical thinking is key to verify the quality and accuracy of AI-generated output, identify biases or assumptions embedded in the AI's responses, ensure relevance and alignment with specific needs, and leverage AI output for strategic goals and appropriate use cases.

While AI can provide a wealth of data and suggestions, the ability to critically evaluate and refine those output is crucial. Without critical thinking, users may accept AI-generated content at face value, risking the perpetuation of errors, biases, or shallow solutions.

2. Creative Thinking and Critical Thinking: A Symbiotic Relationship

Creative thinking and critical thinking are closely intertwined, especially when working with AI:

• Creative Thinking is about generating ideas, exploring possibilities, and approaching problems in novel ways. It's where imagination and innovation flourish.

• Critical Thinking is the systematic approach to analysing, evaluating, and refining those ideas. It's the process of making sure that the creative output is practical, accurate, and aligned with real-world needs.

In the AI context, these two forms of thinking work together to produce outcomes that are both innovative and reliable:

• Creative Thinking helps in designing new prompts, exploring different angles, and generating fresh ideas.

• Critical Thinking ensures that the AI output is accurate, relevant, and free of biases, transforming initial ideas into refined, actionable results.

V. Ethics of AI and Respecting Academic Integrity

A. Understanding Ethical and Professional Boundaries — Three Levels of AI Use

AI use exists on a spectrum — from clear misconduct to responsible use. The challenge lies in identifying the fine line between unethical shortcuts and appropriate, ethical AI integration.

1. Level 1: Prohibited AI Use (Clear Violations of Academic and Professional Integrity)

AI is used in a way that misrepresents work, deceives evaluators, or violates academic/professional policies. This includes plagiarism, falsified data, or misleading automation.

Examples in Academia:

• Submitting AI-generated essays or reports without disclosure.

• Using AI to cheat on exams or generate false citations.

• Copying AI-generated answers for assignments without understanding the material.

Examples in the Workplace:

• Using AI to generate fake reports or fabricate data for clients.

• Relying on AI to write emails or proposals verbatim without review.

• Using AI to automate work without approval, leading to compliance violations.

Why This is a Problem:

• Ethically wrong → It misrepresents skills, knowledge, and effort.

• Legal and reputational risks → Academic penalties, negative impact on career progression, or even lawsuits.

• Loss of credibility → Professors, colleagues, and employers will question trustworthiness.

2. Level 2: Unethical AI Use (Allowed but Ethically Questionable & Professionally Risky)

AI is used in ways that may not explicitly violate rules but undermine learning, creativity, and accountability.

Examples in Academia:

• Over-reliance on AI to summarise readings instead of engaging with the material.

• Asking AI to generate essay structures and arguments without personal analysis.

• Using AI-generated explanations without verifying accuracy.

Examples in the Workplace:

• Letting AI generate client reports or strategic documents without validating insights.

• Using AI to rephrase an entire work document that would not meet company standards.

• Relying on AI for problem-solving without understanding the logic behind recommendations.

Why This is a Problem:

• Diminishes learning and skills development → Over-reliance on AI prevents independent thinking.

• Creates risks of misinformation → AI-generated responses may be inaccurate, incomplete, or biased.

• Lowers work quality → AI cannot fully grasp context, tone, or human nuances.

Key Ethical Question: Would you be comfortable explaining your AI use to a professor, manager, or client?

3. Level 3: Ethical & Responsible AI Use (AI as a Support Tool, Not a Replacement)

AI is used transparently and responsibly to enhance our work, without replacing human thinking, analysis, or ethical judgment.

Examples in Academia:

• Using AI for brainstorming and structuring ideas while ensuring originality.

• Fact-checking AI-generated explanations before including them in research.

• Using AI-powered tools for grammar and clarity improvements in writing.

Examples in the Workplace:

• Using AI to analyse large datasets and extract insights—but personally interpreting findings.

• Leveraging AI for content drafting, but refining tone, structure, and accuracy before publishing.

• Using AI for productivity tasks (e.g., meeting summaries) while maintaining oversight.

Why This is Ethical and Effective:

• Enhances learning and productivity → AI supports, but doesn't replace, thinking.

• Maintains accountability → The final work remains your own.

• Builds credibility → AI use is transparent, reviewed, and aligned with ethical guidelines.

VI. Preparing for an AI-Driven Workforce

A. How Employers Evaluate Talent (Forward-Looking Companies):

• Shift from just degrees to demonstrable skills.

• Actively test candidates on their ability to use AI tools to complete tasks and solve problems.

• AI acts as a performance amplifier, elevating the output of individuals.

B. Top Emerging Skills Employers Demand:

• Creativity & Original Thinking: To solve complex problems uniquely.

• AI Literacy & Prompt Engineering: Effectively working with AI systems, not just querying them.

• Data Literacy & Decision-Making: Understanding problems and testing solutions.

• Ethical & Responsible AI Use: Knowing the boundaries.

C. AI is NOT Just for Coders:

• AI is a versatile tool for brainstorming, content creation (text, image, video), marketing, writing, and more.

• Soon, skill in designing and managing AI agents (bots that automate multi-step processes) will be critical.

D. Changing Career Progression:

• Transition may be rough as companies "right-size" for AI efficiency.

• The era of one education for a lifelong career is over. Continuous upskilling is essential.

• Entrepreneurship is more accessible than ever—AI enables starting a venture with a smaller team.

E. Practical Advice for Students Entering the Workforce:

• Build an AI Portfolio: Create a website or collection of projects (from classes, internships, or personal work) that showcases your applied AI skills.

• Prepare for Skill-Based Interviews: Expect to be tested on using AI creatively and ethically during the interview process itself. Be ready to explain your process.